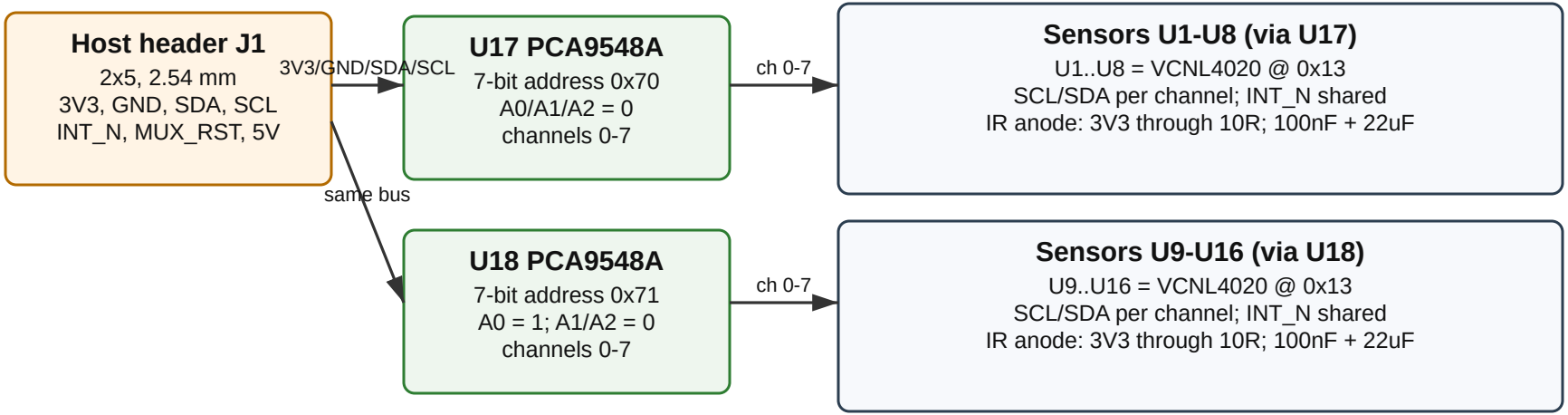
PSM V1 - Electrical Architecture (thesis baseline)

16x Vishay VCNL4020 (fixed 7-bit address 0x13), 2x PCA9548A 8-channel I2C switch, Tiva/LaunchPad header.



Channel map

U17 ch0..7 -> U1..U8 U18 ch0..7 -> U9..U16

Each VCNL4020: 1 IR anode, 2 SDA, 3 INT, 4 SCL, 5 VDD, 8/9 GND, 6/7/10 NC.

Pull-ups: 4.7k on upstream SDA/SCL; 10k on MUX_RST and INT_N. 22uF bulk + 100nF per sensor.

Board: 101.6 x 101.6 mm, 4x4 grid at 25.4 mm pitch, 0.2 mm polyimide flex, 2 layers.